

DELHI PUBLIC SCHOOL

Bokaro Steel City, Jharkhand

PHYSICS — CLASS 11 EXAMINATION

SUBJECT: PHYSICS

TIME ALLOWED: 2.5 HOURS

MAX MARKS: 60

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. All questions are compulsory.
2. Read all instructions carefully before writing.
3. Marks are indicated next to each question.
4. Draw neat and labeled diagrams wherever necessary.
5. Use of log tables is permitted. Use of scientific calculators is prohibited.

SECTION A — MULTIPLE CHOICE QUESTIONS

Select the most appropriate option from the given choices. Each question carries 1 mark.

Q1. Which of the following physical quantities has the same dimensional formula as that of Planck's constant (h)? (1)

- (a) a) Linear momentum (b) b) Angular momentum
(c) c) Work done (d) d) Force

Q2. For a projectile projected from ground level, the maximum horizontal range is achieved when the angle of projection with the horizontal is: (1)

- (a) a) 30 degrees (b) b) 45 degrees
(c) c) 60 degrees (d) d) 90 degrees

Q3. A body of mass 'm' is moving in a circular path of radius 'r' with a constant speed 'v'. The work done by the centripetal force on the body during one complete revolution is: (1)

- (a) a) mv^2/r (b) b) $2\pi r$
(c) c) Zero (d) d) $m \cdot v^2$

Q4. During an adiabatic expansion of an ideal gas from state (P1, V1, T1) to (P2, V2, T2), the work done by the gas is given by: (1)

- (a) a) $R \cdot (T1 - T2) / (2 \cdot \gamma)$ (b) b) $R \cdot (T2 - T1) / (2 \cdot \gamma)$
(c) c) $P1 \cdot V1 \cdot \ln(V2 / V1)$ (d) d) Zero

SECTION B — SHORT QUESTIONS

Answer the following questions in 2 to 4 sentences. Each question carries 2 marks.

Q1. State the law of conservation of linear momentum. Write its mathematical expression for an isolated system containing two colliding bodies. (2)

Q2. Distinguish between conservative and non-conservative forces with one suitable example of each. (2)

Q3. Why does a cyclist bend inwards while taking a sharp turn on a horizontal unbanked road? Explain briefly. (2)

SECTION C — DIAGRAM/GRAPH-BASED QUESTIONS

Explain the concepts and derive formulas based on diagrams, graphical definitions, or physical curves. Each question carries 5 marks.

Q1. A standard Stress vs. Strain curve is plotted for a metal wire under tensile stress. Describe the structural characteristics of this curve by defining and explaining the significance of: (i) Proportional Limit, (ii) Yield Point (Elastic Limit), and (iii) Fracture/Breaking Point. (5)

Q2. Consider a standard velocity-time (v - t) graph representing the motion of a body with uniform acceleration. (a) State what physical quantities are represented by the slope of the curve and the area under the curve. (b) Derive the second kinematic equation of motion, $s = ut + \frac{1}{2}at^2$, using this graphical analysis. (5)

Q3. Draw the Free Body Diagram (FBD) of a block of mass ' m ' sliding down an inclined plane of angle θ with the horizontal, where the coefficient of kinetic friction is μ . Write down the equations of forces parallel and perpendicular to the incline, and derive the expression for the acceleration ' a ' of the block. (5)

Q4. Using a Pressure-Volume (P - V) indicator diagram, trace the thermodynamic stages of a classical Carnot Cycle. Explicitly name each of the four thermodynamic processes involved, stating whether work is done by the gas or on the gas in each stage. (5)

Q5. Graphically analyze the longitudinal standing waves generated in a closed organ pipe (closed at one end, open at the other) of length L . Describe the locations of nodes and antinodes, and derive the ratio of frequencies for the fundamental mode, the second harmonic (first overtone), and the third harmonic (second overtone). (5)

SECTION D — NUMERICAL PROBLEMS

Solve the following numerical problems showing the formula used, step-by-step substitution, and the final answer with appropriate units. Each question carries 5 marks.

Q1. Calculate the value of acceleration due to gravity (g') at a height of 1600 km above the surface of the Earth. Take the radius of the Earth as $R = 6400$ km and the acceleration due to gravity on the Earth's surface as $g = 9.8 \text{ m/s}^2$. (5)

Q2. A circular racetrack of radius 300 m is banked at an angle of 15 degrees. If the coefficient of static friction between the tires of a race car and the track surface is 0.2, calculate: (i) the optimum speed of the car to avoid any wear and tear on its tires, and (ii) the maximum permissible speed of the car to prevent slipping. (Take $g = 9.8 \text{ m/s}^2$, $\tan(15^\circ) = 0.268$). (5)

Q3. A constant force acts on a body of mass 2 kg which is initially at rest. The body acquires a velocity of 10 m/s in a time interval of 4 seconds. Calculate: (i) the total work done by the force on the body, and (ii) the average power delivered by the force during this time interval. (5)

Q4. A particle of mass 0.5 kg executes Simple Harmonic Motion (SHM) with an amplitude of 10 cm and a time period of 2 seconds. Calculate: (i) the maximum velocity of the particle, and (ii) the total energy of the oscillating particle. (Keep your answer in terms of π or compute using $\pi = 3.14$). (5)

Q5. Eight spherical rain droplets of equal size are falling through the atmosphere with a terminal velocity of 5 cm/s. If all eight droplets coalesce to form a single larger drop, what will be the terminal velocity of the new larger drop? (Assume density of air is negligible compared to water and both follow Stokes' law). (5)

ANSWER KEY

SECTION A — MULTIPLE CHOICE QUESTIONS

- Q1.** b) Angular momentum. Explanation: The dimensional formula of Planck's constant (h) is $[M L^2 T^{-1}]$, which is identical to the dimensional formula of Angular Momentum ($L = r \times p$, having dimensions $[L] \times [M L T^{-1}] = [M L^2 T^{-1}]$). [1 mark]
- Q2.** b) 45 degrees. Explanation: The range of a projectile is given by $R = \frac{u^2 \sin(2\theta)}{g}$. The range R is maximum when $\sin(2\theta) = 1$. This occurs when $2\theta = 90^\circ$, $\theta = 45^\circ$. [1 mark]
- Q3.** c) Zero. Explanation: Centripetal force acts towards the center of the circular path, while the instantaneous displacement is along the tangent to the path. Since the angle between force and displacement vectors is 90 degrees at all points, the work done $W = F \cdot d \cdot \cos(90^\circ) = 0$. [1 mark]
- Q4.** a) $R \cdot (T_1 - T_2) / (2 \cdot \theta)$ Explanation: For an adiabatic process, the work done is given by $W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$. Since $PV = RT$ for an ideal gas, this relation simplifies to $W = \frac{R}{\gamma - 1} (T_1 - T_2)$. [1 mark]

SECTION B — SHORT QUESTIONS

- Q1.** The law of conservation of linear momentum states that if no net external force acts on a system, the total linear momentum of the system remains constant. Mathematically, for two colliding bodies of masses m_1 and m_2 with initial velocities u_1 , u_2 and final velocities v_1 , v_2 : $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$. [2 marks]
- Q2.** A force is conservative if the work done by or against it in moving a body between two points is independent of the path taken (e.g., Gravitational force). A force is non-conservative if the work done by it depends on the specific path followed by the body (e.g., Frictional force). [2 marks]
- Q3.** A cyclist bends inwards to produce a horizontal component of the normal reaction force towards the center of the turn. This horizontal component provides the necessary centripetal force for safe turning, preventing the cyclist from slipping outwards due to centrifugal force. [2 marks]

SECTION C — DIAGRAM/GRAPH-BASED QUESTIONS

- Q1.** Model Answer: [5 marks]
- Proportional Limit: This is the region from origin to proportional limit where Hooke's Law is strictly obeyed. Stress is directly proportional to strain, and the graph is a straight line.
 - Yield Point (Elastic Limit): Beyond the proportional limit, the material still behaves elastically but stress-strain linearity is lost. If the load is removed up to this point, the wire will recover its original length without permanent deformation.
 - Plastic Deformation and Ultimate Tensile Strength: Beyond the yield point, the wire undergoes irreversible plastic deformation. Even when load is removed, a permanent strain remains.
 - Fracture/Breaking Point: After reaching the ultimate strength point, necking occurs in the wire, stress increases locally, and the wire finally snaps at the fracture point.
- Q2.** Model Answer: [5 marks]
- (a) Graphical Definitions: The slope of the velocity-time graph (dv/dt) represents acceleration (a), and the total area enclosed between the curve and the time axis represents displacement (s).
- (b) Graphical Derivation:
- Let a body start with an initial velocity ' u ' at $t = 0$ and accelerate uniformly to velocity ' v ' at time ' t '.
 - The area under the v - t straight line from $t = 0$ to t is composed of two parts: a rectangle of height ' u ' and base ' t ', and a right-angled triangle of height ' $(v - u)$ ' and base ' t '.
 - Total Area (Displacement s) = Area of Rectangle + Area of Triangle
 $s = u \cdot t + \frac{1}{2} \cdot t \cdot (v - u)$
 - Since acceleration $a = (v - u) / t$, we can substitute $(v - u) = a \cdot t$.
 - Therefore, $s = u \cdot t + \frac{1}{2} \cdot t \cdot (a \cdot t) = u \cdot t + \frac{1}{2} \cdot a \cdot t^2$.

- Q3.** Model Answer: [5 marks]
- Free Body Diagram Description: Resolve the weight (mg) acting vertically downwards into two components: $mg \cos(\theta)$ perpendicular to the incline, and $mg \sin(\theta)$ down the incline. The Normal force (N) acts perpendicular to the incline (upwards). The kinetic friction force (f_k) acts up the incline (opposing the sliding motion).
 - Equations of equilibrium/motion:
 - Perpendicular to incline: $N = mg \cos(\theta)$
 - Parallel to incline: $mg \sin(\theta) - f_k = ma$
 - Since $f_k = \mu N$, $f_k = \mu mg \cos(\theta)$
 - Substitute f_k into the parallel force equation:
 $mg \sin(\theta) - \mu mg \cos(\theta) = ma$
 - Solving for acceleration 'a' by dividing both sides by 'm':
 $a = g (\sin(\theta) - \mu \cos(\theta))$

- Q4.** Model Answer: [5 marks]
- A Carnot cycle consists of four successive processes on a P-V diagram:
- Stage 1 to 2: Isothermal Expansion. The gas absorbs heat Q_1 from the source at constant temperature T_1 . The volume increases from V_1 to V_2 . Work is done BY the gas ($W_1 > 0$).
 - Stage 2 to 3: Adiabatic Expansion. The gas is thermally insulated and expands further from volume V_2 to V_3 , dropping its temperature from T_1 to T_2 . Work is done BY the gas ($W_2 > 0$).
 - Stage 3 to 4: Isothermal Compression. The gas is brought into contact with a cold sink at temperature T_2 . It is compressed from volume V_3 to V_4 , releasing heat Q_2 . Work is done ON the gas ($W_3 < 0$).
 - Stage 4 to 1: Adiabatic Compression. The gas is insulated and compressed back from V_4 to its initial volume V_1 , rising its temperature back to T_1 . Work is done ON the gas ($W_4 < 0$).

- Q5.** Model Answer: [5 marks]
- Wave behavior: In a closed organ pipe, a displacement node (zero air movement) always forms at the closed end, and a displacement antinode (maximum air movement) forms at the open end.
 - Fundamental Mode (First Harmonic):
 - Wave configuration: One node at closed end, one antinode at open end.
 - Length relation: $L = \frac{1}{4} \lambda \Rightarrow \lambda = 4L$
 - Frequency: $f_1 = v / \lambda = v / 4L$
 - First Overtone (Third Harmonic):
 - Wave configuration: Node - Antinode - Node - Antinode. Two nodes and two antinodes.
 - Length relation: $L = \frac{3}{4} \lambda \Rightarrow \lambda = \frac{4L}{3}$
 - Frequency: $f_3 = v / \lambda = 3v / 4L = 3f_1$ (Even harmonics are absent in closed pipes, so this is the 3rd Harmonic).
 - Second Overtone (Fifth Harmonic):
 - Wave configuration: Three nodes and three antinodes.
 - Length relation: $L = \frac{5}{4} \lambda \Rightarrow \lambda = \frac{4L}{5}$
 - Frequency: $f_5 = v / \lambda = 5v / 4L = 5f_1$ (monic).
 - Ratio of Frequencies: The ratio of successive frequencies generated in a closed organ pipe is $f_1 : f_3 : f_5 = 1 : 3 : 5$. (Only odd harmonics are present).

SECTION D — NUMERICAL PROBLEMS

- Q1.** Formula: [5 marks]
- $$g' = g \left[\frac{R}{R + h} \right]^2$$
- Given:
- Acceleration on surface (g) = 9.8 m/s^2
 - Earth's Radius (R) = 6400 km
 - Height (h) = 1600 km
- Substitution:
- Calculate total distance from center: $R + h = 6400 \text{ km} + 1600 \text{ km} = 8000 \text{ km}$.
 - Apply formula: $g' = 9.8 \times \left[\frac{6400}{8000} \right]^2$
 - Simplify the fraction: $\frac{6400}{8000} = \frac{64}{80} = \frac{4}{5} = 0.8$
 - Square the simplified fraction: $(0.8)^2 = 0.64$
 - Calculate g' : $g' = 9.8 \times 0.64 = 6.272 \text{ m/s}^2$.
- Final Answer:
The acceleration due to gravity at a height of 1600 km is 6.27 m/s^2 .

Q2.

Formula:

[5 marks]

(i) Optimum speed: $v_{\text{opt}} = \sqrt{r * g * \tan(\theta)}$ (ii) Maximum speed: $v_{\text{max}} = \sqrt{[r * g * (\frac{1}{\mu} + \tan(\theta))] / (1 - \mu \tan(\theta))}$

Given:

- Radius (r) = 300 m

- Angle (θ) = 15° - $\tan(15^\circ) = 0.268$ - Coefficient of friction (μ) = 0.2- $g = 9.8 \text{ m/s}^2$

Substitution:

(i) Optimum Speed calculation:

$$v_{\text{opt}} = \sqrt{300 * 9.8 * 0.268}$$

$$v_{\text{opt}} = \sqrt{787.92} = 28.07 \text{ m/s.}$$

(ii) Maximum Speed calculation:

$$v_{\text{max}} = \sqrt{[300 * 9.8 * (0.2 + 0.268) / (1 - 0.2 * 0.268)]}$$

$$v_{\text{max}} = \sqrt{2940 * (0.468) / (1 - 0.0536)}$$

$$v_{\text{max}} = \sqrt{1375.92 / 0.9464}$$

$$v_{\text{max}} = \sqrt{1453.84} = 38.13 \text{ m/s.}$$

Final Answer:

(i) The optimum speed of the car is 28.07 m/s.

(ii) The maximum permissible speed is 38.13 m/s.

Q3.

Formula:

[5 marks]

(i) Work Done (by Work-Energy Theorem): $W = \frac{1}{2} * m * (v^2 - u^2)$ (ii) Average Power: $P = W / t$

Given:

- Mass (m) = 2 kg

- Initial velocity (u) = 0 m/s

- Final velocity (v) = 10 m/s

- Time (t) = 4 s

Substitution:

(i) Work Done calculation:

$$W = \frac{1}{2} * 2 * (10^2 - 0^2)$$

$$W = 1 * (100)$$

$$W = 100 \text{ Joules.}$$

(ii) Average Power calculation:

$$P = 100 / 4$$

$$P = 25 \text{ Watts.}$$

Final Answer:

(i) The total work done by the force is 100 J.

(ii) The average power delivered is 25 W.

Q4.

Formula:

[5 marks]

- Angular frequency (ω) = $2\pi / T$ - (i) Maximum Velocity: $v_{\text{max}} = \omega * A$ - (ii) Total Energy: $E = \frac{1}{2} * m * \omega^2 * A^2$

Given:

- Mass (m) = 0.5 kg

- Amplitude (A) = 10 cm = 0.1 m

- Time period (T) = 2 s

Substitution:

First, find ω

$$\omega = 2\pi / T = 2\pi / 2 = \pi \text{ rad/s} \approx 3.14 \text{ rad/s.}$$

(i) Maximum Velocity:

$$v_{\text{max}} = \omega * A$$

$$v_{\text{max}} = 3.14 * 0.1 = 0.314 \text{ m/s.}$$

(ii) Total Energy:

$$E = \frac{1}{2} \times 0.5 \times \left(\frac{2\pi}{T} \right)^2 \times r^2$$

$$E = 0.25 \times 9.87 \times 0.01 \text{ (using } \pi = 3.14 \text{)}$$

$$E = 0.0247 \text{ Joules.}$$

Final Answer:

(i) The maximum velocity of the particle is 0.314 m/s.

(ii) The total energy of the oscillating particle is 0.0247 J (or 0.025 J).

Q5.

Formula:

[5 marks]

$$\text{Terminal Velocity: } v = \frac{2 \times r^2 \times (\rho - \sigma) \times g}{9 \times \eta}$$

This implies terminal velocity is directly proportional to the square of the radius: $v \propto r^2$.

Given:

- Terminal velocity of small droplet (v_1) = 5 cm/s

- Number of droplets (n) = 8

Substitution:

1. Let r be the radius of a small droplet, and R be the radius of the coalesced big drop.

2. Volume conservation: Volume of big drop = 8 * Volume of small droplet

$$\frac{4}{3} \pi R^3 = 8 \times \frac{4}{3} \pi r^3$$

$$R^3 = 8 \times r^3 \Rightarrow R = 2 \times r$$

3. Set up the terminal velocity proportion:

$$v_2 / v_1 = (R / r)^2$$

$$v_2 / 5 = (2 \times r / r)^2$$

$$v_2 / 5 = (2)^2$$

$$v_2 / 5 = 4$$

$$v_2 = 5 \times 4 = 20 \text{ cm/s.}$$

Final Answer:

The terminal velocity of the new larger drop will be 20 cm/s.